The EBM model reads

$$C(\vec{x})\frac{\partial T}{\partial t} + A + BT = \underbrace{\nabla\cdot(D(\vec{x})\nabla T)}_{\mathcal{L}} + \underbrace{QS(\vec{x},t)a}_{\hat{S}}$$

In spherical coordinates on the sphere's surface (longitude, latitude) = (λ, θ)

$$\mathcal{L} = \csc^2(\theta)\frac{\partial}{\partial\lambda}\left(D\frac{\partial T}{\partial\lambda}\right) + \frac{\partial}{\partial\theta}\left(D\frac{\partial T}{\partial\theta}\right) + \cot(\theta)\, D\frac{\partial T}{\partial\theta}$$

When we apply the FD/FV method, we have for each DOF

$$\left(\frac{\partial T}{\partial t}(t=\tau)\right)_{ij} = \left(\frac{\partial T}{\partial t}\right)^{\tau}_{ij} \approx \frac{1}{C_{ij}}\left[\hat{S}^{\tau}_{ij} - A - BT_{ij} + \mathcal{L}_{ij}\right]$$

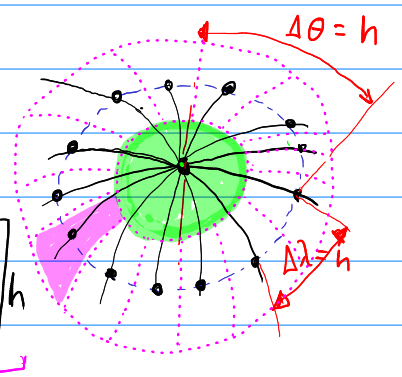
The discretization of the heat conduction term on the surface of a sphere with a Cartesian mesh of cell size h (uniform for latitude & longitude) reads:

- For the inner DOFs:

$$\mathcal{L}_{ij} = \csc^2(\theta_j)\frac{1}{h^2}\left[-2D_{ij}T_{ij} + \left[D_{ij} - \frac{1}{4}\left(D_{(i+1,j)} - D_{(i-1,j)}\right)\right]T_{(i-1,j)} + \left[D_{ij} + \frac{1}{4}\left(D_{(i+1,j)} - D_{(i-1,j)}\right)\right]T_{(i+1,j)}\right]$$

$$+ \frac{1}{h^2}\left[-2D_{ij}T_{ij} + \left[D_{ij} - \frac{1}{4}\left(D_{(i,j+1)} - D_{(i,j-1)}\right)\right]T_{(i,j-1)} + \left[D_{ij} + \frac{1}{4}\left(D_{(i,j+1)} - D_{(i,j-1)}\right)\right]T_{(i,j+1)}\right]$$

$$+ \cot(\theta_j)\frac{D_{ij}}{2h}\left[T_{(i,j+1)} - T_{(i,j-1)}\right]$$



- For the DOFs at the poles:

$$\mathcal{L}_{ij} = \frac{\sin(h/2)}{\text{area}[1]}\sum_{k=1}^{n_x}\left[\frac{D_{k,1}\,\text{area}[1] + D_{k,2}\,\text{area}[2]}{\text{area}[1] + \text{area}[2]}\right]\left[\frac{T_{k,2} - T_{k,1}}{h}\right]h$$

where $\text{area}[1] = 0{,}5\left(1 - \cos\left(\frac{h}{2}\right)\right)$ contains the normalized area of the polar cap and

$\text{area}[2] = \frac{\sin\left(\frac{h}{2}\right)\sin(h)}{n_x}$ contains the area of a cell in the second row of nodes

The Jacobian matrix of the operator $F(T) = \frac{\partial T}{\partial t}$ is defined as

$$J = \text{reshape}\left(\frac{\partial F}{\partial T}, n_x \times n_y, n_x \times n_y\right) \in \mathbb{R}^{n_x \times n_y, n_x \times n_y}$$

and can be computed using a finite difference. For example, the row k of the matrix is obtained as

$$J_k = \text{vec}\left(F(v^{i,j}) - F(0)\right) \quad \forall \; i = 1, \dots, n_x, \; j = 1, \dots, n_y$$

where $k = i + n_x(j-1)$ corresponds to a DOF, and $v^{i,j}$ is a unit vector of the size of T, such that

$$v^{i,j}_{l,m} = \begin{cases} 1 & \text{if } i = l, j = m, \\ 0 & \text{otherwise} \end{cases}$$